

Comparitive Performance Report for tentacle-ec-2+2-16k-balance-off-rerun3 vs direct-reads-ec-2+2-16k-balance-off- rerun-pr2

Table of contents

- Comparison summary for tentacle-ec-2+2-16k-balance-off-rerun3 vs direct-reads-ec-2+2-16k-balance-off-rerun-pr2
- Response Curves
 - Sequential Read
 - Random Read
 - Random Read/Write
- Configuration yaml files
 - results

Comparison summary for tentacle-ec-2+2-16k-balance-off-rerun3 vs direct-reads-ec-2+2-16k-balance-off- rerun-pr2

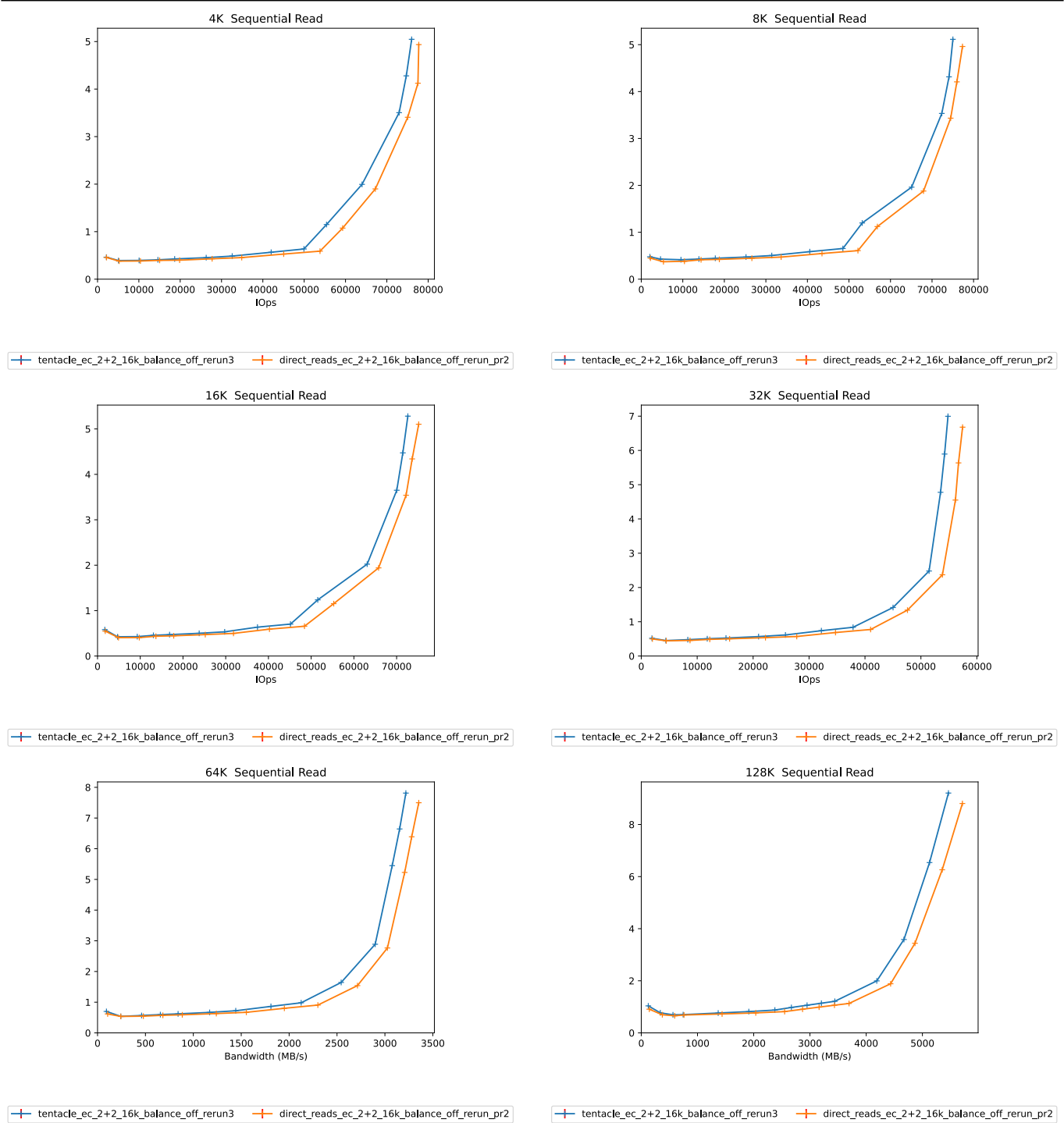
Sequential Read	tentacle_ec_2+2_16k_balance_off_rerun3	direct_reads_ec_2+2_16k_balance_off_rerun_pr2	%change	latency
4K	76008 IOps@5.0ms	77727 IOps@4.9ms	2%	-2%
8K	75022 IOps@5.1ms	77329 IOps@5.0ms	3%	-2%
16K	72629 IOps@5.3ms	75168 IOps@5.1ms	3%	-4%
32K	54841 IOps@7.0ms	57447 IOps@6.7ms	5%	-4%
64K	3219 MB/s@7.8ms	3353 MB/s@7.5ms	4%	-4%
128K	5463 MB/s@9.2ms	5711 MB/s@8.8ms	5%	-4%
256K	7825 MB/s@12.9ms	8330 MB/s@12.1ms	6%	-6%
384K	8704 MB/s@23.1ms	9469 MB/s@21.3ms	9%	-8%
512K	9283 MB/s@28.9ms	10213 MB/s@26.3ms	10%	-9%
768K	9350 MB/s@43.1ms	10354 MB/s@38.9ms	11%	-10%
1024K	9496 MB/s@42.4ms	10544 MB/s@38.2ms	11%	-10%
2048K	8973 MB/s@14.9ms	9725 MB/s@27.6ms	8%	85%
4096K	8490 MB/s@31.6ms	8989 MB/s@89.6ms	6%	184%

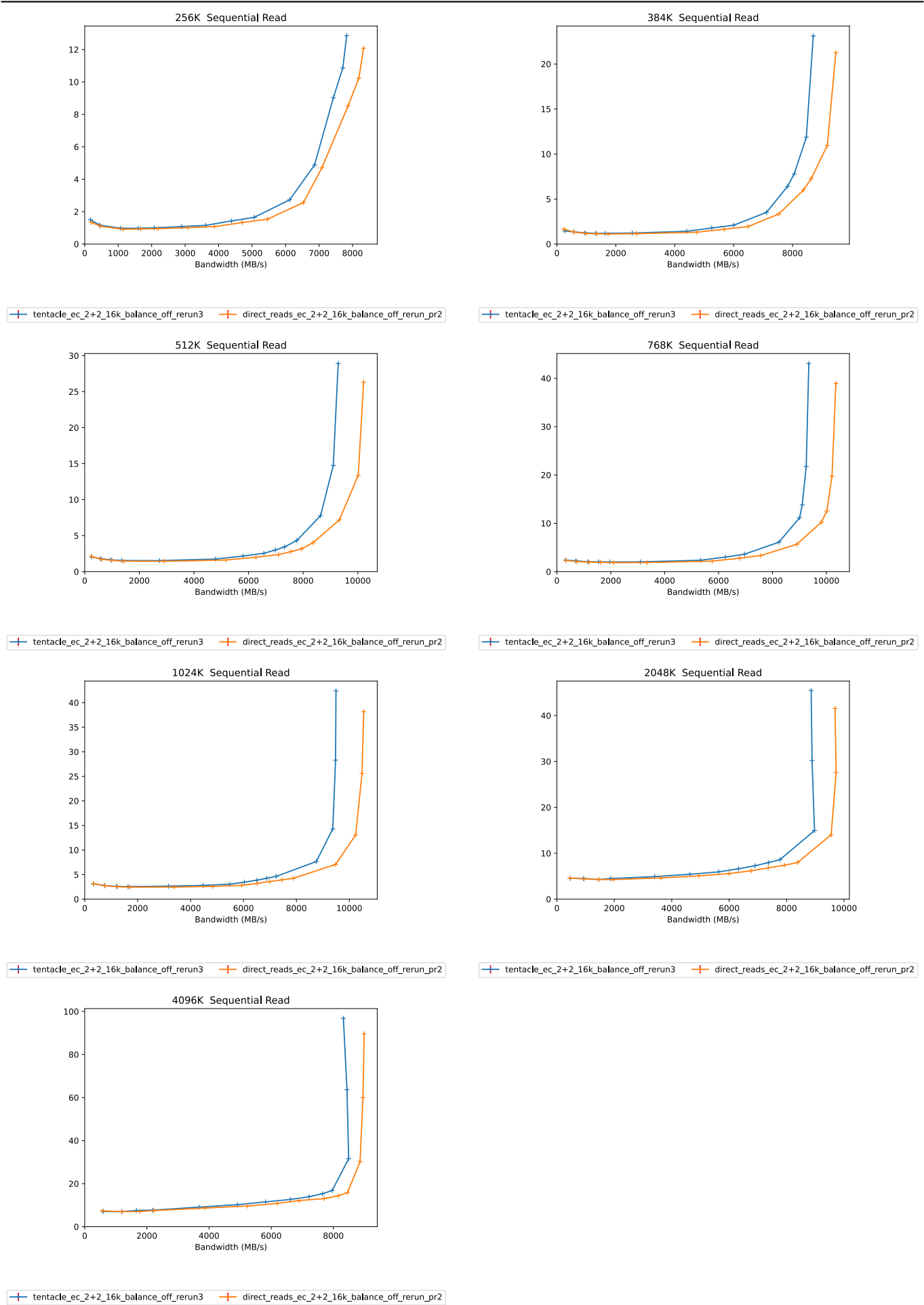
Random Read	tentacle_ec_2+2_16k_balance_off_rerun3	direct_reads_ec_2+2_16k_balance_off_rerun_pr2	%change	latency
4K	73319 IOps@5.2ms	75857 IOps@5.1ms	3%	-2%
8K	71092 IOps@5.4ms	73165 IOps@4.4ms	3%	-19%
16K	68554 IOps@5.6ms	70725 IOps@5.4ms	3%	-4%
32K	48537 IOps@7.9ms	49626 IOps@7.7ms	2%	-3%
64K	2811 MB/s@8.9ms	2890 MB/s@8.7ms	3%	-2%
128K	4789 MB/s@10.5ms	4996 MB/s@10.1ms	4%	-4%
256K	7312 MB/s@9.2ms	7745 MB/s@13.0ms	6%	41%
384K	7964 MB/s@6.3ms	8947 MB/s@11.2ms	12%	78%
512K	8703 MB/s@30.8ms	9687 MB/s@27.7ms	11%	-10%
768K	8941 MB/s@45.0ms	9963 MB/s@40.5ms	11%	-10%
1024K	9146 MB/s@44.0ms	10142 MB/s@39.7ms	11%	-10%
2048K	8839 MB/s@30.4ms	9438 MB/s@28.4ms	7%	-7%
4096K	8447 MB/s@31.8ms	8985 MB/s@59.8ms	6%	88%

Random Read/Write	tentacle_ec_2+2_16k_balance_off_rerun3	direct_reads_ec_2+2_16k_balance_off_rerun_pr2	%change	latency
4K_70/30	36665 IOps@10.5ms	37441 IOps@8.5ms	2%	-19%
16K_70/30	34355 IOps@11.2ms	35734 IOps@10.7ms	4%	-4%
64K_70/30	1699 MB/s@14.8ms	1766 MB/s@11.9ms	4%	-20%
64K_30/70	1221 MB/s@13.7ms	1274 MB/s@19.7ms	4%	44%

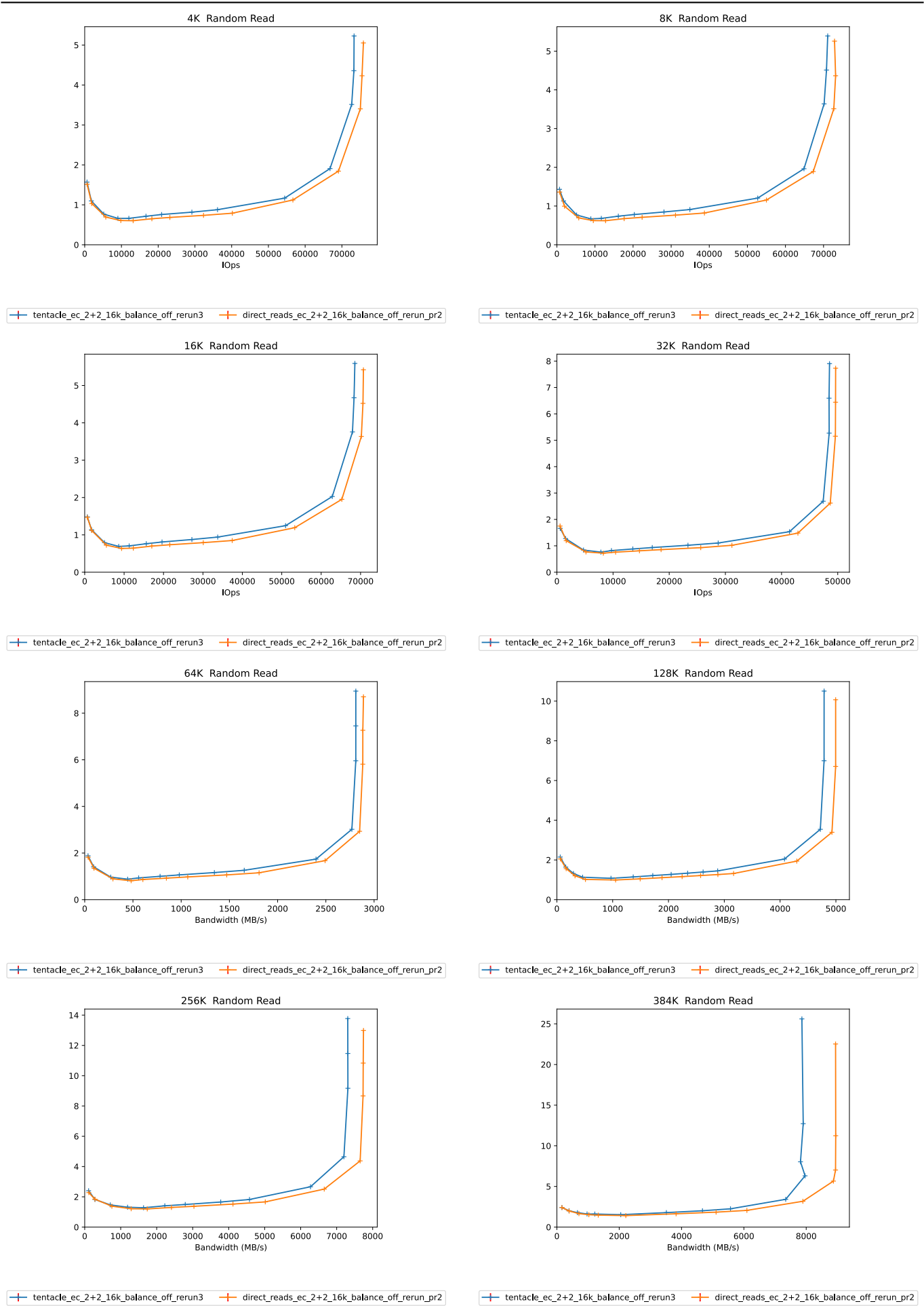
Response Curves

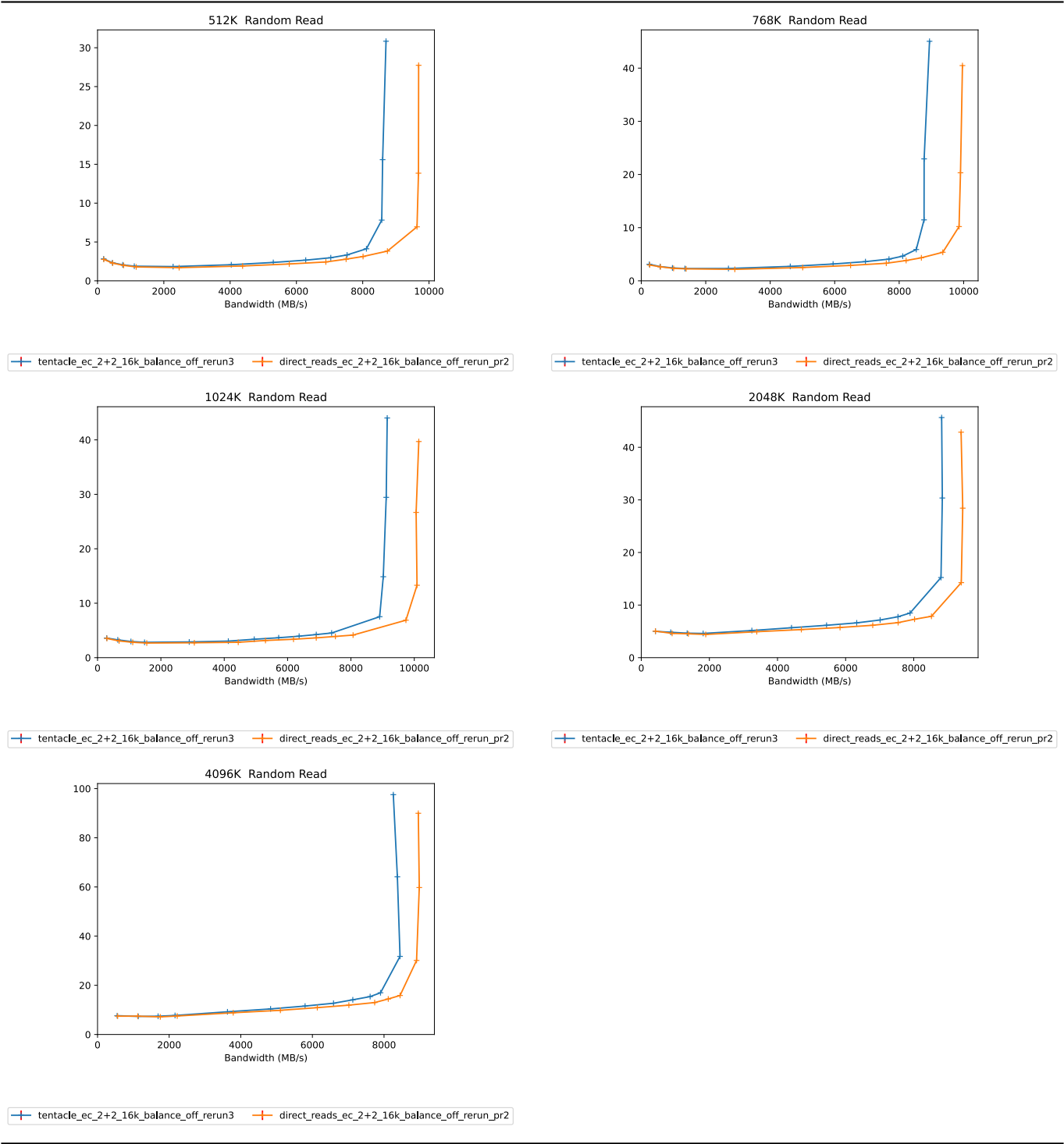
Sequential Read



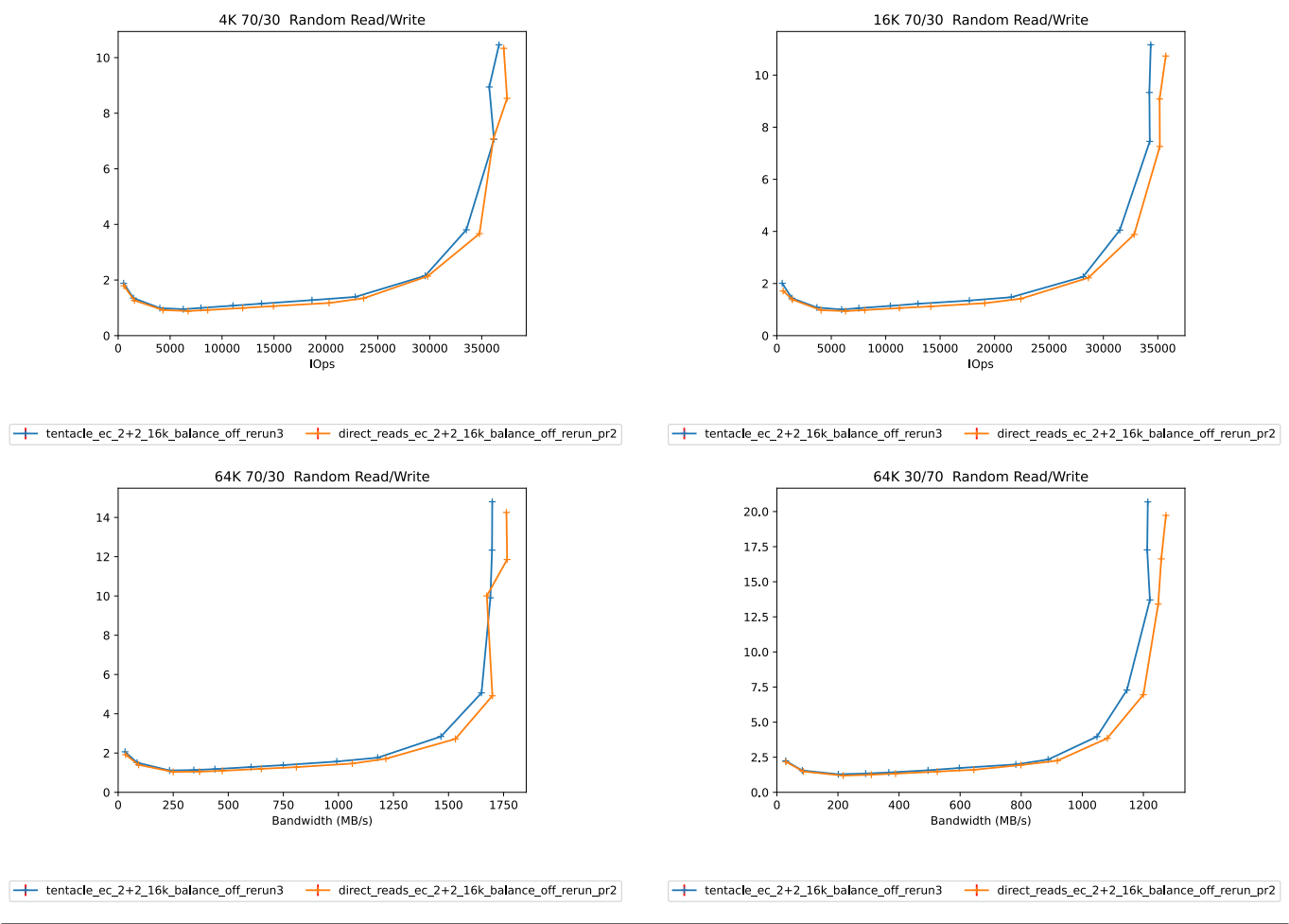


Random Read





Random Read/Write



Configuration yaml files

Only yaml files that differ by more than 20 lines from the yaml file for the baseline directory will be added here in addition to the baseline yaml

results

```

librbd fio:
  cmd_path: /usr/local/bin/fio2
  create_report: true
  fio_out_format: json
  log_avg_msec: 100
  log_bw: true
  log_iops: true
  log_lat: true
  norandommap: true
  osd_ra:
    - 4096
  poolname: rbd_replicated
  prefill:
    blocksize: 64k
    numjobs: 1
  procs_per_volume:
    - 1
  ramp: 30
  rbdname: cbt-librbd fio
  time: 90
  time_based: true
  use_existing_volumes: true
  vol_size: 1000
  volumes_per_client:
    - 16
  wait_pgautoscaler_timeout: 10
  workloads:
    128k randomread:
      jobname: randread
      mode: randread
      numjobs:
        - 1
      op_size: 131072
      total_iodepth:
        - 1
        - 2
        - 3
        - 4
        - 8
        - 12
        - 16
        - 20
        - 24
        - 28
        - 32
        - 64
        - 128
        - 256
        - 384
    128k sequentialread:
      jobname: seqread
      mode: read
      numjobs:
        - 1
      op_size: 131072
      total_iodepth:
        - 1
        - 2
        - 3
        - 4
        - 8

```

```
- 12
- 16
- 20
- 24
- 28
- 32
- 64
- 128
- 256
- 384
16kmixread70:
  jobname: mixread
  mode: randrw
  numjobs:
    - 1
  op_size: 16384
  rwmixread: 70
  total_iodepth:
    - 1
    - 2
    - 4
    - 6
    - 8
    - 12
    - 16
    - 24
    - 32
    - 64
    - 128
    - 256
    - 320
    - 384
16krandomread:
  jobname: randread
  mode: randread
  numjobs:
    - 1
  op_size: 16384
  total_iodepth:
    - 1
    - 2
    - 4
    - 6
    - 8
    - 12
    - 16
    - 24
    - 32
    - 64
    - 128
    - 256
    - 320
    - 384
16ksequentialread:
  jobname: seqread
  mode: read
  numjobs:
    - 1
  op_size: 16384
  total_iodepth:
    - 1
    - 2
    - 4
```

```

- 6
- 8
- 12
- 16
- 24
- 32
- 64
- 128
- 256
- 320
- 384
1Mrandomread:
  jobname: randread
  mode: randread
  numjobs:
    - 1
  op_size: 1048576
  total_iodepth:
    - 1
    - 2
    - 3
    - 4
    - 8
    - 12
    - 16
    - 20
    - 24
    - 28
    - 32
    - 64
    - 128
    - 256
    - 384
1Msequentialread:
  jobname: seqread
  mode: read
  numjobs:
    - 1
  op_size: 1048576
  total_iodepth:
    - 1
    - 2
    - 3
    - 4
    - 8
    - 12
    - 16
    - 20
    - 24
    - 28
    - 32
    - 64
    - 128
    - 256
    - 384
256krandomread:
  jobname: randomread
  mode: randread
  numjobs:
    - 1
  op_size: 262144
  total_iodepth:
    - 1

```

```

- 2
- 4
- 6
- 8
- 12
- 16
- 24
- 32
- 64
- 128
- 256
- 320
- 384
256ksequentialread:
  jobname: seqread
  mode: read
  numjobs:
    - 1
  op_size: 262144
  total_iodepth:
    - 1
    - 2
    - 4
    - 6
    - 8
    - 12
    - 16
    - 24
    - 32
    - 64
    - 128
    - 256
    - 320
    - 384
2Mrandomread:
  jobname: randread
  mode: randread
  numjobs:
    - 1
  op_size: 2097152
  total_iodepth:
    - 1
    - 2
    - 3
    - 4
    - 8
    - 12
    - 16
    - 20
    - 24
    - 28
    - 32
    - 64
    - 128
    - 192
2Msequentialread:
  jobname: seqread
  mode: read
  numjobs:
    - 1
  op_size: 2097152
  total_iodepth:
    - 1

```

- 2
- 3
- 4
- 8
- 12
- 16
- 20
- 24
- 28
- 32
- 64
- 128
- 192

32krandomread:

- jobname: randread
- mode: randread
- numjobs:
- 1
- op_size: 32768
- total_iodepth:
- 1
- 2
- 4
- 6
- 8
- 12
- 16
- 24
- 32
- 64
- 128
- 256
- 320
- 384

32ksequentialread:

- jobname: seqread
- mode: read
- numjobs:
- 1
- op_size: 32768
- total_iodepth:
- 1
- 2
- 4
- 6
- 8
- 12
- 16
- 24
- 32
- 64
- 128
- 256
- 320
- 384

384krandomread:

- jobname: randread
- mode: randread
- numjobs:
- 1
- op_size: 393216
- total_iodepth:
- 1

```

- 2
- 3
- 4
- 5
- 8
- 16
- 24
- 32
- 64
- 128
- 160
- 256
- 512
384ksequentialread:
  jobname: seqread
  mode: read
  numjobs:
    - 1
  op_size: 393216
  total_iodepth:
    - 1
    - 2
    - 3
    - 4
    - 5
    - 8
    - 16
    - 24
    - 32
    - 64
    - 128
    - 160
    - 256
    - 512
4Mrandomread:
  jobname: randread
  mode: randread
  numjobs:
    - 1
  op_size: 4194304
  total_iodepth:
    - 1
    - 2
    - 3
    - 4
    - 8
    - 12
    - 16
    - 20
    - 24
    - 28
    - 32
    - 64
    - 128
    - 192
4Msequentialread:
  jobname: seqread
  mode: read
  numjobs:
    - 1
  op_size: 4194304
  total_iodepth:
    - 1

```

```
- 2
- 3
- 4
- 8
- 12
- 16
- 20
- 24
- 28
- 32
- 64
- 128
- 192
4kmixread70:
  jobname: mixread
  mode: randrw
  numjobs:
    - 1
  op_size: 4096
  rwmixread: 70
  total_iodepth:
    - 1
    - 2
    - 4
    - 6
    - 8
    - 12
    - 16
    - 24
    - 32
    - 64
    - 128
    - 256
    - 320
    - 384
4krandomread:
  jobname: randread
  mode: randread
  numjobs:
    - 1
  op_size: 4096
  total_iodepth:
    - 1
    - 2
    - 4
    - 6
    - 8
    - 12
    - 16
    - 24
    - 32
    - 64
    - 128
    - 256
    - 320
    - 384
4ksequentialread:
  jobname: seqread
  mode: read
  numjobs:
    - 1
  op_size: 4096
  total_iodepth:
```


- 1
- 2
- 4
- 6
- 8
- 12
- 16
- 24
- 32
- 64
- 128
- 256
- 320
- 384

512krandomread:

- jobname: randread
- mode: randread
- numjobs:
- 1
- op_size: 524288
- total_iodepth:
- 1
- 2
- 3
- 4
- 8
- 16
- 24
- 32
- 40
- 48
- 64
- 128
- 256
- 512

512ksequentialread:

- jobname: seqread
- mode: read
- numjobs:
- 1
- op_size: 524288
- total_iodepth:
- 1
- 2
- 3
- 4
- 8
- 16
- 24
- 32
- 40
- 48
- 64
- 128
- 256
- 512

64kmixread30:

- jobname: mixread
- mode: randrw
- numjobs:
- 1
- op_size: 65536
- rwmixread: 30

```
total_iodepth:
- 1
- 2
- 4
- 6
- 8
- 12
- 16
- 24
- 32
- 64
- 128
- 256
- 320
- 384
64kmixread70:
  jobname: mixread
  mode: randrw
  numjobs:
  - 1
  op_size: 65536
  rwmixread: 70
  total_iodepth:
  - 1
  - 2
  - 4
  - 6
  - 8
  - 12
  - 16
  - 24
  - 32
  - 64
  - 128
  - 256
  - 320
  - 384
64krandomread:
  jobname: randread
  mode: randread
  numjobs:
  - 1
  op_size: 65536
  total_iodepth:
  - 1
  - 2
  - 4
  - 6
  - 8
  - 12
  - 16
  - 24
  - 32
  - 64
  - 128
  - 256
  - 320
  - 384
64ksequentialread:
  jobname: seqread
  mode: read
  numjobs:
  - 1
```

```
  op_size: 65536
  total_iodepth:
  - 1
  - 2
  - 4
  - 6
  - 8
  - 12
  - 16
  - 24
  - 32
  - 64
  - 128
  - 256
  - 320
  - 384
768krandomread:
  jobname: randread
  mode: randread
  numjobs:
  - 1
  op_size: 786432
  total_iodepth:
  - 1
  - 2
  - 3
  - 4
  - 8
  - 16
  - 24
  - 32
  - 40
  - 48
  - 64
  - 128
  - 256
  - 512
768ksequentialread:
  jobname: seqread
  mode: read
  numjobs:
  - 1
  op_size: 786432
  total_iodepth:
  - 1
  - 2
  - 3
  - 4
  - 5
  - 8
  - 16
  - 24
  - 32
  - 64
  - 128
  - 160
  - 256
  - 512
8krandomread:
  jobname: randread
  mode: randread
  numjobs:
  - 1
```

```

    op_size: 8192
    total_iodepth:
      - 1
      - 2
      - 4
      - 6
      - 8
      - 12
      - 16
      - 24
      - 32
      - 64
      - 128
      - 256
      - 320
      - 384
  8ksequentialread:
    jobname: seqread
    mode: read
    numjobs:
      - 1
    op_size: 8192
    total_iodepth:
      - 1
      - 2
      - 4
      - 6
      - 8
      - 12
      - 16
      - 24
      - 32
      - 64
      - 128
      - 256
      - 320
      - 384
  precondition:
    jobname: precond1rw
    mode: randwrite
    monitor: false
    numjobs:
      - 1
    op_size: 65536
    time: 600
    total_iodepth:
      - 16
cluster:
  archive_dir: /tmp/cbt
  ceph-mgr_cmd: /usr/bin/ceph-mgr
  ceph-mon_cmd: /usr/bin/ceph-mon
  ceph-osd_cmd: /usr/bin/ceph-osd
  ceph-run_cmd: /usr/bin/ceph-run
  ceph_cmd: /usr/bin/ceph
  clients:
    - --- server1 ---
  clusterid: ceph
  conf_file: /cbt/ceph.conf.4x1x1.fs
  fs: xfs
  head: --- server1 ---
  iterations: 1
  mgrs:
    --- server1 ---:

```

```
    a: null
mkfs_opts: -f -i size=2048
mons:
  --- server1 ---:
    a: --- IP Address ---:6789
mount_opts: -o inode64,noatime,logbsize=256k
osds:
- --- server1 ---
  osds_per_node: 6
  pdsh_ssh_args: -a -x -l%u %h
  rados_cmd: /usr/bin/rados
  rbd_cmd: /usr/bin/rbd
  tmp_dir: /tmp/cbt
  use_existing: true
  user: cjames
monitoring_profiles:
  collectl:
    args: -c 18 -sCD -i 10 -P -oz -FO --rawtoo --sep ";" -f {collectl_dir}
```